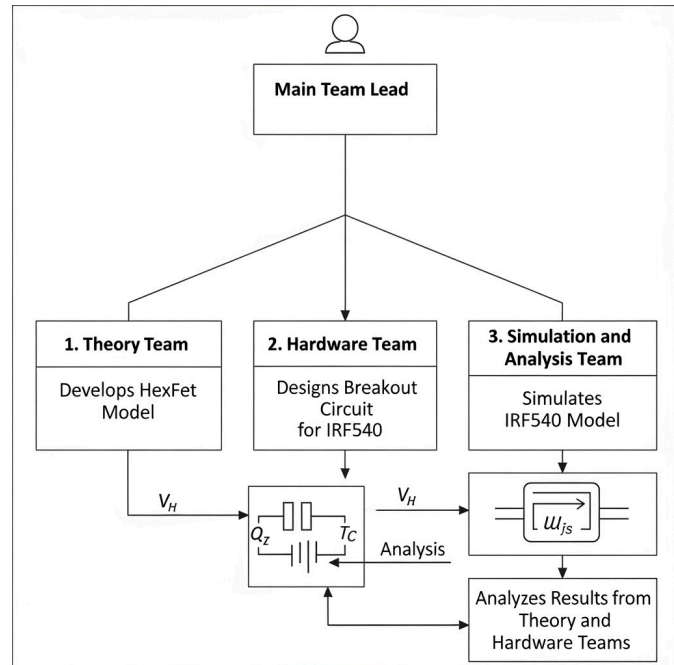


WorkFlow structure for week 3:

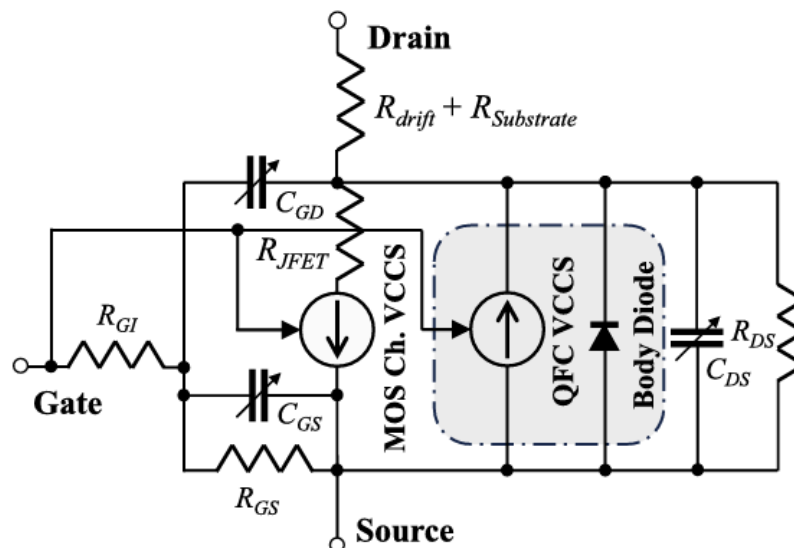
We divided our team into almost 3 equal three equal sub-teams to handle the tasks of

1. Developing the HexFet Model using the Quasi_Floating Channel Assumption, delegated to the theory team.
2. Designing a breakout circuit for the IRF540 to collect the real world values and behavior of the IRF540 in a, the Forward Bias conditioning and B, Reverse Bias or the third quadrant region.
3. Simulation the model of the IRF540 and analysing the results from the hardware and the theory teams, delegated to the simulation and analysis team.



Theoretical Analysis:

Below is the methodology we followed to determine the parameters of the IRF540 as used in the Quasi-Floating channel model assumption. The following data and parameters are derived using the IRF540 datasheet provided by Infineon and their Hexfet model.



Parameters to be determined:

R_{GI}: It represents the resistance of the polysilicon gate structure itself. While it's a small value, it works in conjunction with the input capacitance to affect the switching speed. The datasheet's switching time tests are performed with a total gate resistance (R_G on IRF540N Datasheet) of **5.1Ω**, which includes this internal resistance plus the external resistor used in the test circuit

R_{GS}: This represents the leakage path across the gate oxide. We can calculate its equivalent value from the Gate-to-Source Leakage Current (I_{GSS}). The datasheet specifies a maximum I_{GSS} of **100 nA** at V_{GS} = **20V**, Using Ohm's Law: $R = VI$,

$$\text{so, } R_{GS} = 20 \text{ V} / 100 \text{ nA} = \mathbf{200M\Omega}$$

This very high resistance indicates an extremely low leakage current, which is characteristic of a healthy gate oxide

R_{DSon} (R_{drift} + R_{substrate}) : These two resistances, along with the channel resistance and contact resistances, combine to form the total Static Drain-to-Source On-Resistance **R_{DSon}**. The datasheet does not separate these individual contributions, but it provides their total sum. For the IRF540N, the maximum specified **R_{DSon}** is **44 mΩ** when the device is fully turned on (V_{GS}=10V). This is the most critical resistance for calculating conduction losses

(A lower RDS(on) means less power loss and better efficiency,)

R_{DS}: This represents the leakage path when the MOSFET is in the "off" state. Its value can be calculated from the Drain-to-Source Leakage Current (I_{DSS}). The datasheet specifies a maximum I_{DSS} of 25 μA at V_{DS}=100V (when V_{GS} = 0V).

$$\mathbf{R_{DS} = 100V / 25 \mu A = 4M\Omega}$$

Parasitic Capacitances:

The internal capacitances are not fixed values; they are highly dependent on the applied voltage, specifically the drain-to-source voltage (V_{DS}). The datasheet provides composite values (C_{iss}, C_{oss}, C_{rss}), from which we can derive the individual capacitances.

The relationships:

$$C_{GD} = C_{rss}$$

$$C_{GS} = C_{iss} - C_{rss}$$

$$C_{DS} = C_{oss} - C_{rss}$$

Values for V_{DS} = 25V;

$$C_{iss} \text{ (Input Capacitance)} = 1960 \text{ pF}$$

$$C_{oss} \text{ (Output Capacitance)} = 250 \text{ pF}$$

$$C_{rss} \text{ (Reverse Transfer "Miller" Capacitance)} = 40 \text{ pF}$$

The calculated values:

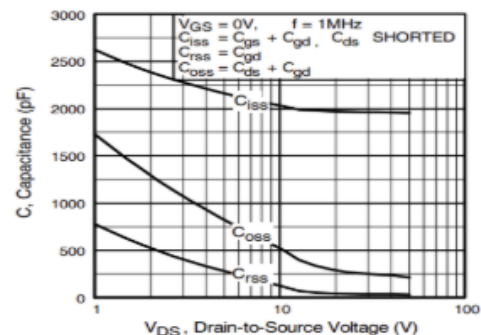


Fig 5. Typical Capacitance Vs. Drain-to-Source Voltage

$$C_{GD} = 40 \text{ pF}$$

$$C_{GS} = 1960 \text{ pF} - 40 \text{ pF} = 1920 \text{ pF}$$

$$C_{DS} = 250 \text{ pF} - 40 \text{ pF} = 210 \text{ pF}$$

As shown in Figure 5 of the datasheet, these capacitances vary significantly with V_{DS} . This is because the depletion region of the internal body diode widens as the reverse voltage (V_{DS}) across it increases, effectively pushing the "plates" of the capacitors further apart.

Here is the approximate range of values, estimated from the graph in Figure 5:

Range of C_{GD} : ~100pF @ $V_{DS} \sim 1V$

Range of C_{GS} : ~2700pF @ $V_{DS} \sim 1V$

Range of C_{DS} : ~400pF @ $V_{DS} \sim 1V$

Range of C_{GD} : ~5pF @ $V_{DS} \sim 100V$

Range of C_{GS} : ~795pF @ $V_{DS} \sim 100V$

Range of C_{DS} : ~45pF @ $V_{DS} \sim 100V$

Table of the determined parameters:

Parameter	Value
R_{GI}	5.1Ω
R_{GS}	$200M\Omega$
$R_{DSon} (R_{drift} + R_{substrate})$	$44 \text{ m}\Omega$
R_{DS}	$4M\Omega$
C_{GD}	(~100 to ~5) pF
C_{GS}	(~2700 to ~795) pF
C_{DS}	(~400 to ~45) pF

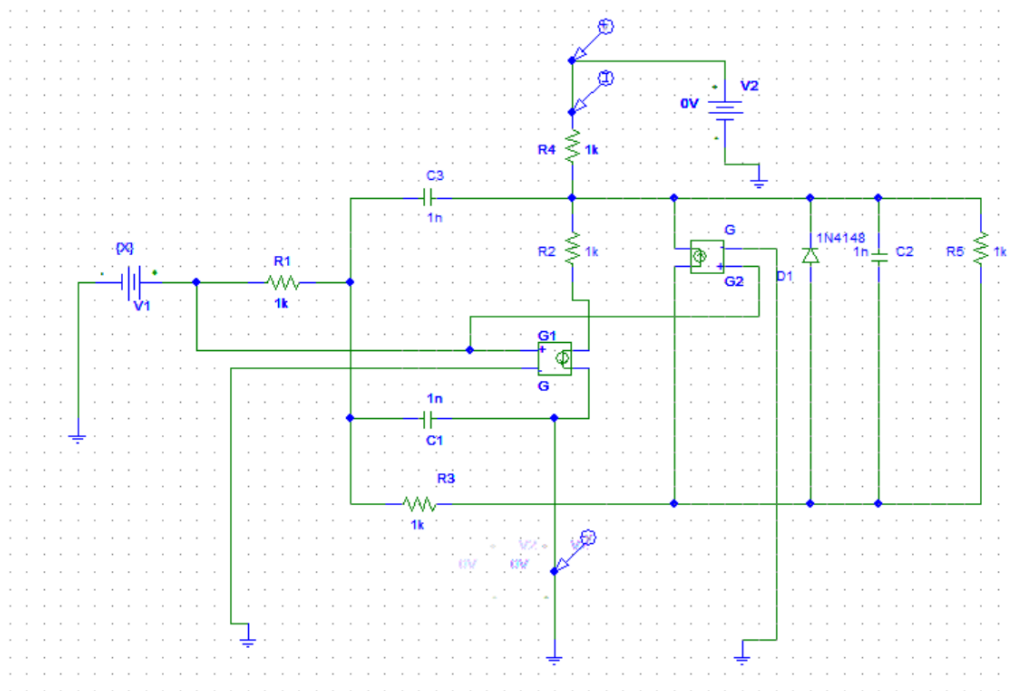
Problems faced:

The theory team faced immense pressure as the task of developing the HexFet model required prior understanding of the subject. Due to this lack of experience and familiarity the accuracy of our calculations may be subjected to further scrutiny.

Take aways:

We must put a bigger emphasis on the theoretical sub-team allocating some more resources to the team for smoother workflow and for skill development conduct tutorials, classes and consultation sessions with the research supervisor.

Simulation of the Quasi Model in Pspice-

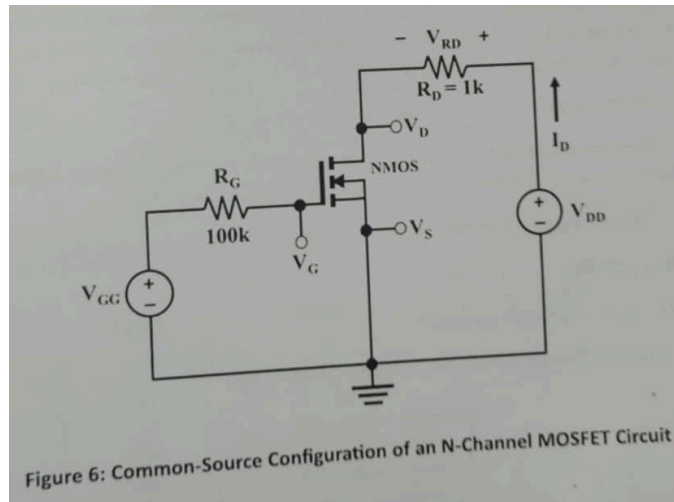


Again, this simulation did not yield the desired output results with the given parameter values.

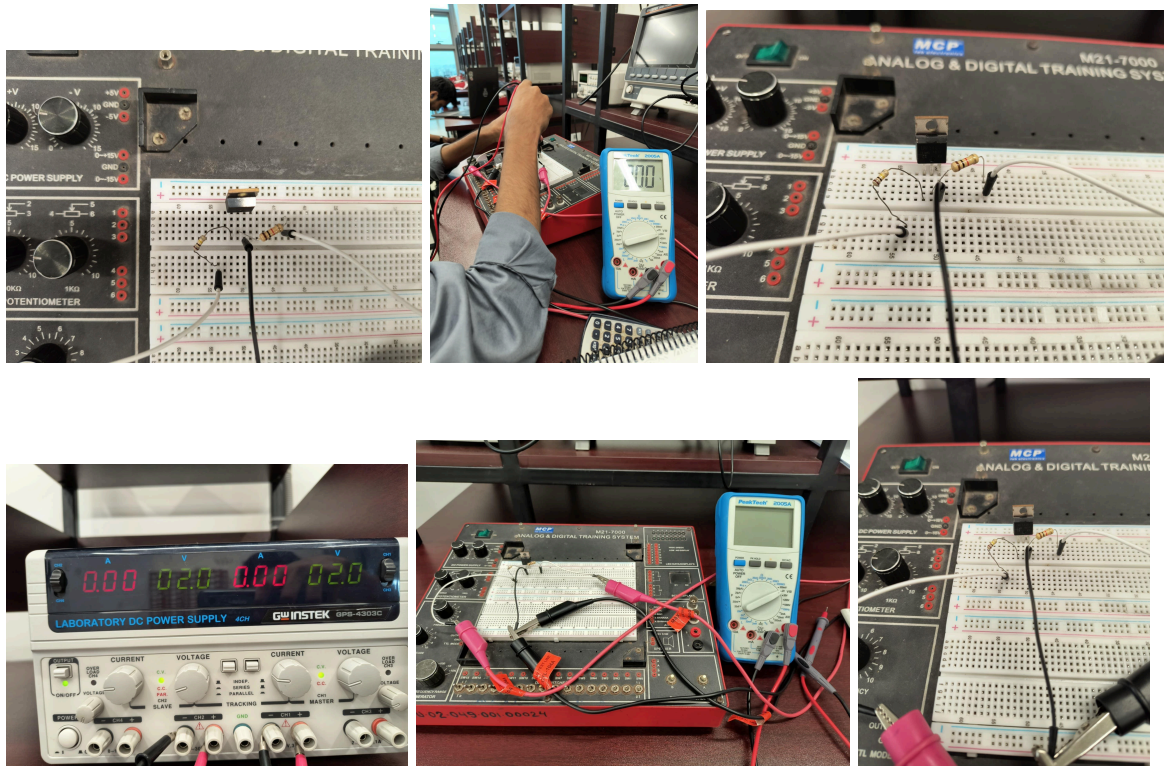
Hardware Experiment Results:

- 1) IRF540 V_{ds} vs I_{ds} curve for different V_{gs}

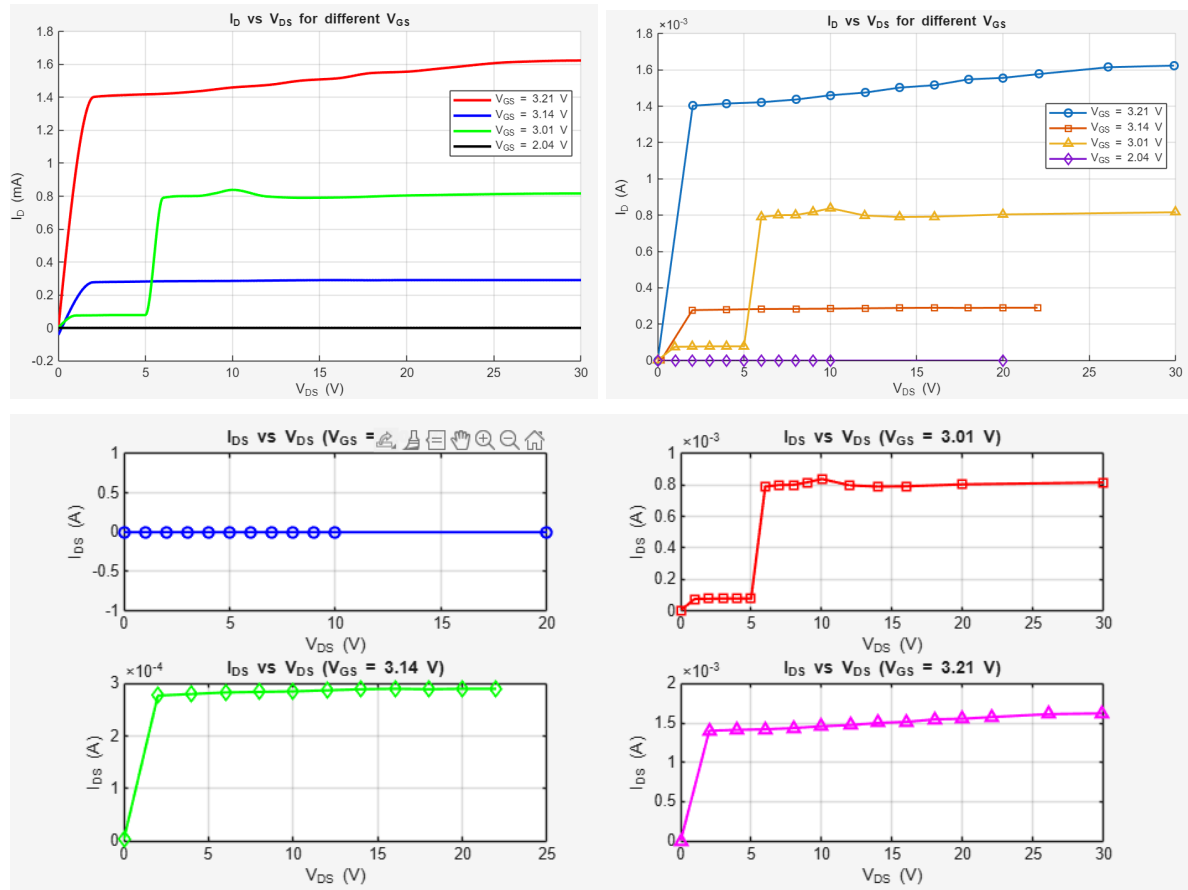
Circuit Diagram-



Setup-



Results-

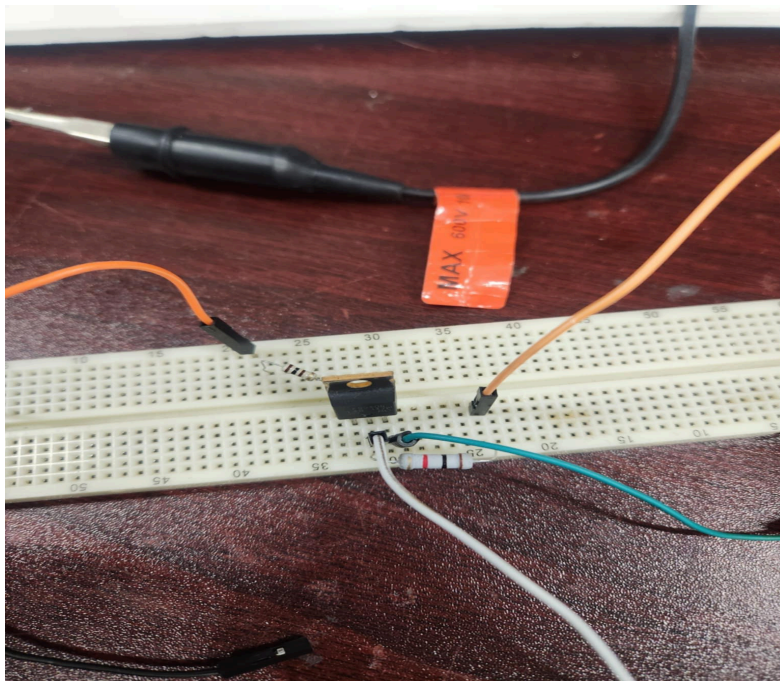
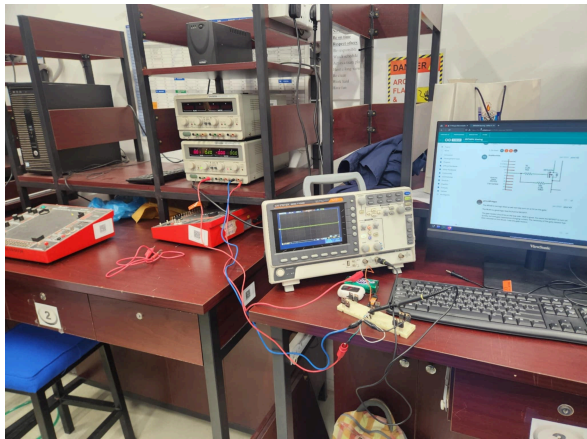


We encountered several issues during our lab work in obtaining the desired results. Our task was to record I_D vs V_{DS} for different V_{GS} values. We found out that our MOSFET has a V_{th} of 3.1V. When we increase V_{GS} beyond that, we can't properly observe the saturation region because our DC source provides only 30V and pushing V_{GS} above ~ 4 V causes the resistors to overheat and burn out due to high current.

We managed to get a few IV curves for lower V_{GS} values. The reference paper used a V_{GS} range from 0 to 18V, which seems difficult with our practical setup.

2) IRF540 V_{ds} vs I_{ds} Graph for reverse bias

Setup-



Results-

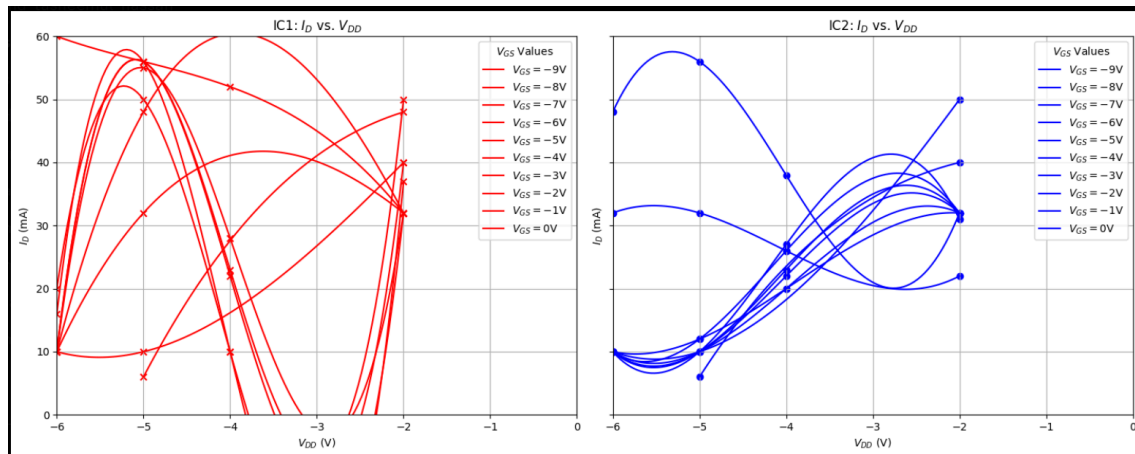


Fig. Reverse Bias Curves for V_{GS} -9 to 0. (Replicating Fig-8 of the “Quasi” Paper)

Here again there have been some anomalies in the graphs for certain values of V_{GS} . Otherwise the curves have come as expected. We used two different ICs of IRF540 to measure the values. IC2 gives us the most optimum results. IC1 has 10% anomalies.

Problems faced:

The IRF540 collected from the lab may have some faults as they were also used by prior teams in their projects. This explains the 10% abnormalities of the average data points in each condition specially in the third quadrant region.

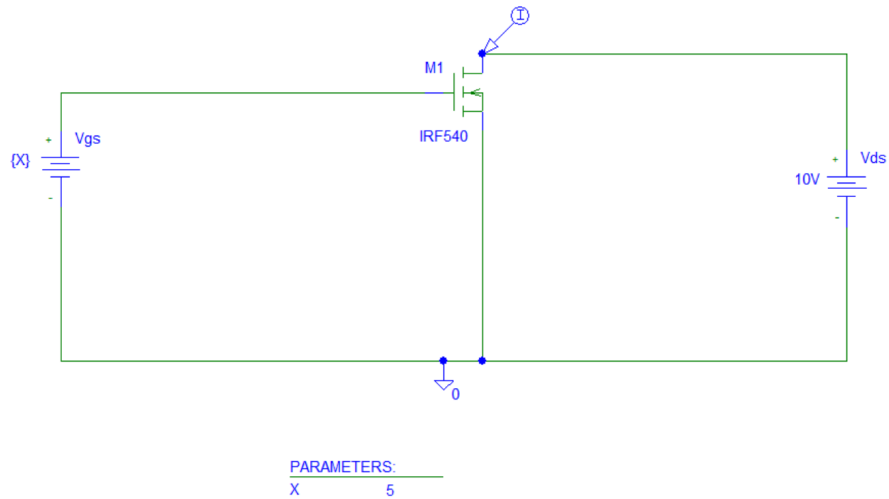
Take aways:

We should build more robust test benches to test the ICs and keep heat sinks in the design instead of using breadboard setup as they are proving to be unreliable and a bit difficult to troubleshoot while dealing with large quantities of data.

Simulation Results:

1) IRF540 Vds vs Ids curve for different Vgs

Circuit Diagram-



Results-

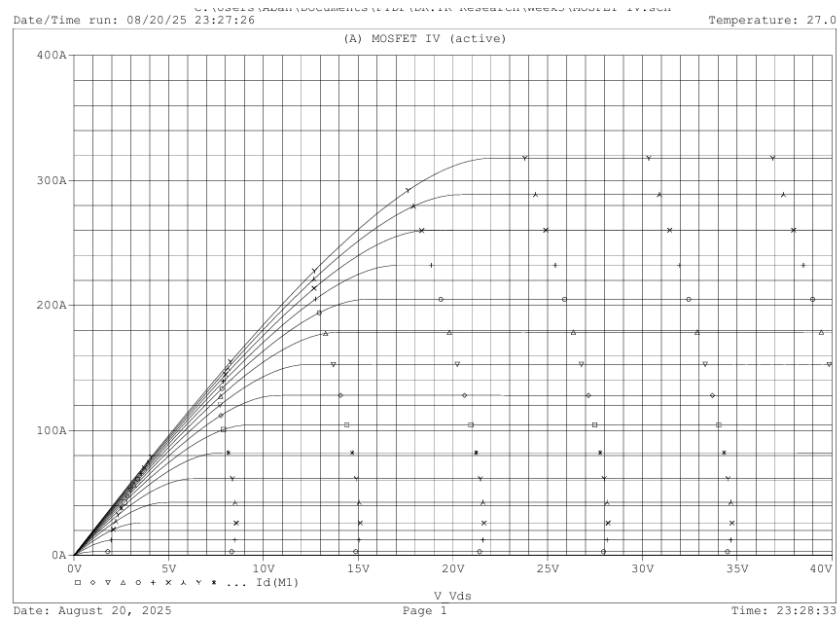


Fig. IV curve for Vgs 0 to 18 (Replicating Fig-6 of the “Quasi” Paper)

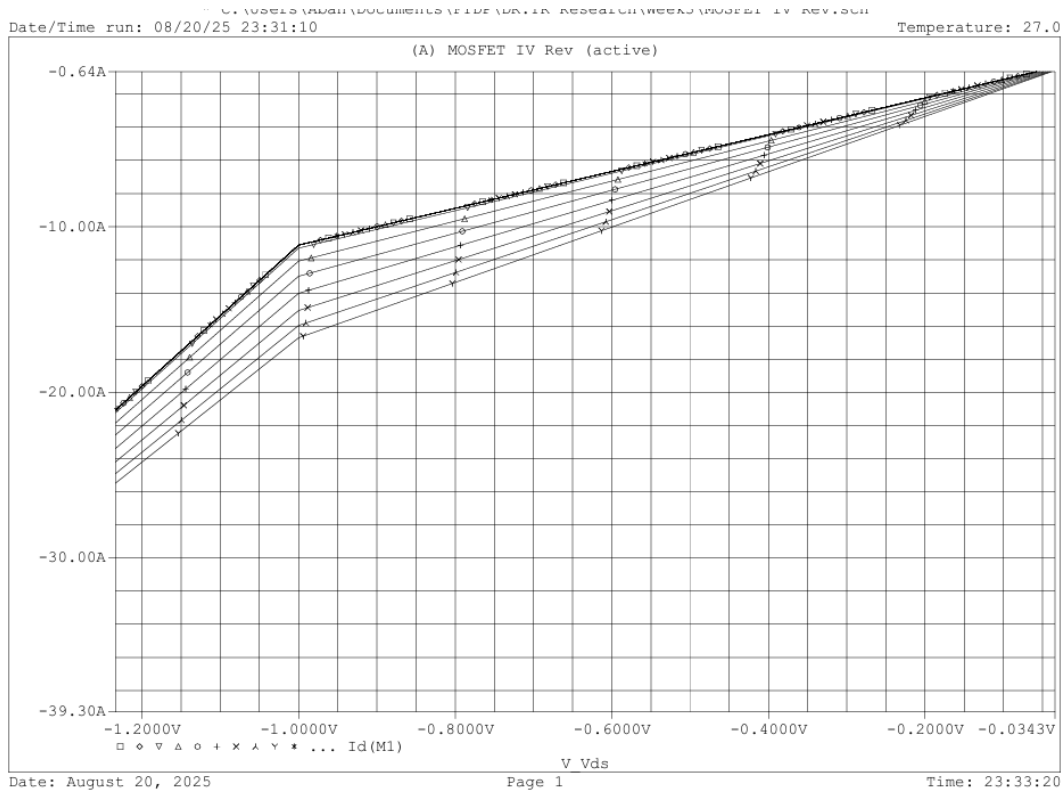


Fig. IV Curve for Reverse Bias (Replicating Fig-8 of the “Quasi” Paper)

In the software simulation, these are the basic simulations that we have been able to replicate for the IRF540. Due to the limitations in finding the theoretical values matching the Quasi State we were unable to simulate the RC Distribution network given in our “Quasi” reference paper.

Reference:

1. Infineon IRF540 datasheet [Website](#)
2. F. -J. Hsu, T. -F. Chang, C. -T. Yen and C. -F. Huang, "A Quasi-Floating Channel-Based SPICE Model for Improving the Modeling Accuracy of SiC MOSFETs With Multiple Device Structures," in IEEE Transactions on Power Electronics, vol. 39, no. 11, pp. 14670-14679, Nov. 2024, doi: 10.1109/TPEL.2024.3433370.